Unit 3 · Lesson 10 Electron Configurations II

Ions, shorthand, valence, Lewis dots Name _____ Per _____

Keywords: ion, cation, anion, noble-gas shorthand, core electrons, valence electrons, Lewis dots, isoelectronic. *Pencil. Fix wrong answers in a second color. Printed text is yours to read; every blank is a decision.*

Question of the day

Do before the bell rings; revisit at the end

Why doesn't your lunch blow up? Circle one: **A** It isn't really potassium. **B** The potassium already lost an electron. **C** Too little of it to react. **D** The peel keeps the water out. **(B is the best answer)**

My reason: _____ After the lesson I would change to: _____

Retrieve

5 min, no notes

1. Config of P (Z = 15): $1s^2 2s^2 2p^6 3s^2 3p^3$ **2.** Electrons in a d subshell: **10** **3.** 17 p^+, 18 e^-. Charge: $1-$

Predict, then watch

commit before the animation

Argon has 18 electrons: $1s^2 2s^2 2p^6 3s^2 3p^6$. Potassium has one more. Electron 19 goes to (circle): **A** 3d **B** 4s **C** 4p. Because _____

After watching: electron 19 went to **4s**; the outermost shell holds **1** electron.

Ions and shorthand

An **ion** is an atom that has gained or lost electrons. **Protons never change**; electrons leave from, and join, the **outermost shell** (the highest *n*). A **cation** lost electrons (positive). An **anion** gained electrons (negative). **Shorthand** swaps the core for the noble gas that comes **before** the element: Na = [Ne]$3s^1$, not [Ar]$3s^1$.

K^+: protons **19** electrons **18** charge **1+** Shorthand for Al (Z = 13): $[Ne]3s^2 3p^1$

Core vs valence: gallium, Z = 31

predict first: how many valence electrons does gallium have? My guess: **3**

Valence electrons are in the outermost shell (highest *n*). Everything inside is **core**. The 3d fills after 4s, but its shell number is 3.

Shell *n*	Subshells in that shell	Electrons	Circle one
1	$1s^2$	**2**	core valence
2	$2s^2 2p^6$	**8**	core valence
3	$3s^2 3p^6 3d^{10}$	**18**	core valence
4	$4s^2 4p^1$	**3**	core valence

Ga has **3** valence electrons.

Check: do the shell counts add to 31? **2+8+18+3**

Lewis dots

atoms only: no brackets or charges yet

Dots show **valence electrons only**. Place the s pair on **top**, then one dot each on the **right, bottom, left**, then pair up. Na and Mg are drawn; do Al to Ar. P should get **5** dots.



1 valence



2 valence



3 valence



4 valence



5 valence



6 valence



7 valence



8 valence

Stretch

finished early? start here

Stretch 1 Draw the orbital boxes for N (Z = 7). Unpaired electrons: **3** Full configuration of Ca: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

Stretch 2 S^{2-}, Cl^-, K^+, Ca^{2+} and Ar share one configuration. Protons in each: **16**, **17**, **19**, **20**, **18**. What differs? **protons**

Stretch 3 Fe is [Ar]$3d^6 4s^2$. Write Fe^{2+}: $[Ar]3d^6$ and Fe^{3+}: $[Ar]3d^5$ Which electrons left first? **4s**

Worked examples: I do, we do, you do

3 marks each

I do: Mg^{2+}

1. Atom has 12 electrons: [Ne]$3s^2$
2. Lose 2 from $n = 3$:
[Ne]$3s^2 \rightarrow$ [Ne]
3. Same as neon: Mg^{2+} = [Ne]

Protons stay 12.

We do: Cl^-

1. Electrons in the atom: **17**
2. Gain 1: [Ne]$3s^2 3p$ **6**
3. Matches: [**Ar**]

Cl^- = [Ar]

You do: S^{2-}

1. Electrons: **18**
2. Full: $1s^2 2s^2 2p^6 3s^2 3p^6$
3. Shorthand: [Ar]

Two ways to get 18. Which did you use?

Pair trade

write your four, swap, check with the table

Do the four ions in your letter's rows. Then check your partner's four. Disagree? Find the **first** line where you differ.

	Ion	Electrons	Shorthand	Partner OK?
A	N^{3-}	**10**	[Ne]	☐
A	Li^+	**2**	[He]	☐
A	K^+	**18**	[Ar]	☐
A	Br^-	**36**	[Kr]	☐
B	Al^{3+}	**10**	[Ne]	☐
B	O^{2-}	**10**	[Ne]	☐
B	Ca^{2+}	**18**	[Ar]	☐
B	Se^{2-}	**36**	[Kr]	☐

Br^- and Se^{2-} hide the $3d^{10}$ trap: is the 3d inside the bracket?

Error hunt

name the flaw, then fix it

	Claim	What is wrong?	Fix
A	Na = [Ar]$3s^1$	**Argon comes after sodium; the noble gas before it is neon.**	[Ne]$3s^1$
B	Ca^{2+} = [Ar]$4s^2$	**That is the atom. Both 4s electrons are gone.**	[Ar]
C	Gallium has 13 valence electrons.	**Counted the 3d (and core). Valence is only $n = 4$.**	**$4s^2 + 4p^1 = 3$**
D	Sulfur's Lewis symbol has 16 dots.	**16 is the atomic number. Dots are valence only.**	**6 dots**

Hinge: arsenic, [Ar]$3d^{10}4s^2 4p^3$

whiteboards up on three

How many valence electrons? **A** 3 **B** 5 **C** 15 **D** 33 My board: ____ Right answer: **B** If I missed it, my error was: _____

Open questions

no wrong thinking here: argue it

Do Lewis dots work for d electrons? What breaks?

Dots track only s and p: 2 + 6 = 8. A d subshell holds 10 more, so the octet has no room. Fe forms Fe^{2+} and Fe^{3+}, so "valence" is unclear.

4s fills before 3d. Why does it leave first?

Once 3d electrons are present, repulsion and the larger shell make the 4s electrons the outermost, so they go first. (Left open on purpose.)

Exit ticket

tear off and hand in

1. Ca^{2+} shorthand: [Ar] **2.** One particle with the same configuration: **Ar, K^+, Cl^-, S^{2-} or P^{3-}**

3. Draw the Lewis dots for phosphorus in the box. Valence electrons: **5**



5 valence